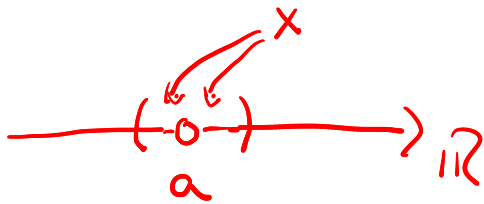


$$a \in \mathbb{R}, L \in \mathbb{R}$$

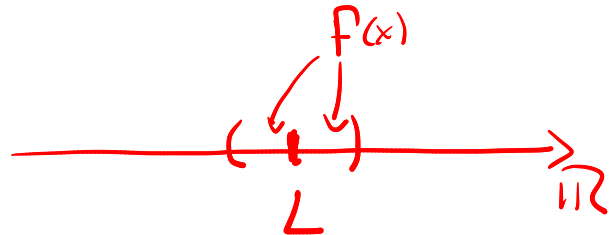
$$\lim_{x \rightarrow a} f(x) = L$$

INTUITIVAMENTE
SIGNIFICA:

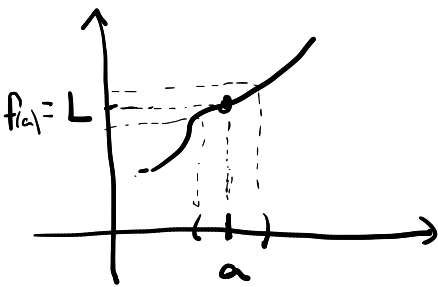
Se x está "próximo" de a ,
então $f(x)$ estará "próximo" de
 L



$\Rightarrow$

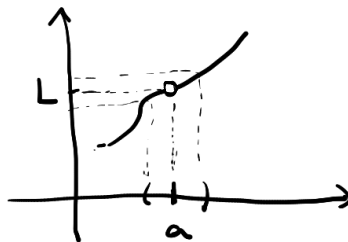


EXEMPLOS:



$$\lim_{x \rightarrow a} f(x) = L$$

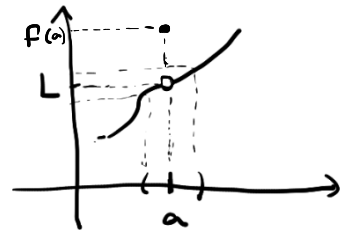
$$a \in D_f$$



$$\lim_{x \rightarrow a} f(x) = L$$

$$a \notin D_f$$

$$\nexists f(a)$$



$$\lim_{x \rightarrow a} f(x) = L$$

$$a \in D_f$$

f é contínua

em a

$$\left(\begin{array}{l} a \in D_f \\ \lim_{x \rightarrow a} f(x) = f(a) \end{array} \right)$$

O que significa "estar próximo"? $\Rightarrow$ CONCEITO FORMAL
(δ, ε)

VOCÊ ESCOLHE

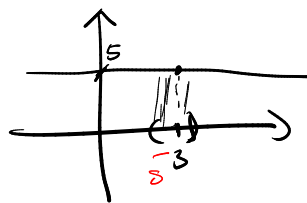
$$\lim_{x \rightarrow a} f(x) = L \Leftrightarrow \text{Dado } \varepsilon > 0, \text{ existe } \delta > 0 \text{ tal que para todo } x \text{ com } 0 < |x - a| < \delta \text{ tem-se } |f(x) - L| < \varepsilon$$

depende do ε dado

Exercícios 2.2

7 $\lim_{x \rightarrow 3} \overset{f(x)}{5} = 5$

$f(x) = 5$



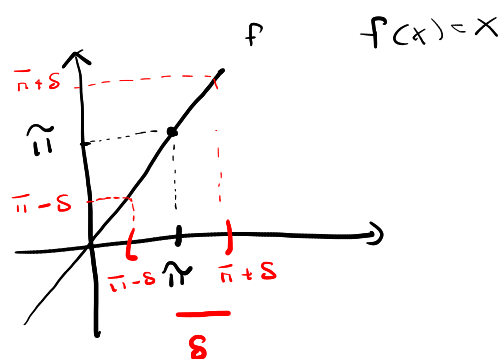
$$|f(x) - 5| < \varepsilon \Rightarrow |5 - 5| < \varepsilon \Rightarrow |0| < \varepsilon$$

Dado $\underline{\varepsilon} > 0$, considere $\delta = \varepsilon$.

$$0 < |x - 3| < \delta \Rightarrow \underbrace{|f(x) - 5|}_{0} < \varepsilon$$

Logo, $\lim_{x \rightarrow 3} f(x) = 5$

9 $\lim_{x \rightarrow \pi} \overset{f(x)}{x} = \pi$



$$|f(x) - \pi| < \varepsilon$$

$\Leftrightarrow$

$$|x - \pi| < \underbrace{\varepsilon}_{\delta}$$

Dado $\varepsilon > 0$, considere $\delta = \varepsilon$.

ENTÃO, se $(x \in D_f) \wedge (0 < |x - \pi| < \varepsilon)$ então,

Como $\underline{f(x) = x}$ teremos $0 < \underbrace{|f(x) - \pi|}_{\varepsilon} < \varepsilon$.

Ou seja, $|f(x) - \pi| < \varepsilon$. Portanto, $\lim_{x \rightarrow \pi} f(x) = \pi$

15

$$f(x) = \begin{cases} x+3, & \text{se } x \neq 3 \\ 7, & \text{se } x = 3 \end{cases}$$

